



جامعة قاسيون الخاصة

مقرر الدارات الكهربائية ١ **Electric Circuits 1**

الدكتور المهندس
عمار كنعان

المحاضرة الرابعة - مسائل

السنة الثانية - هندسة

Mesh Analysis

• تحليل الحلقات

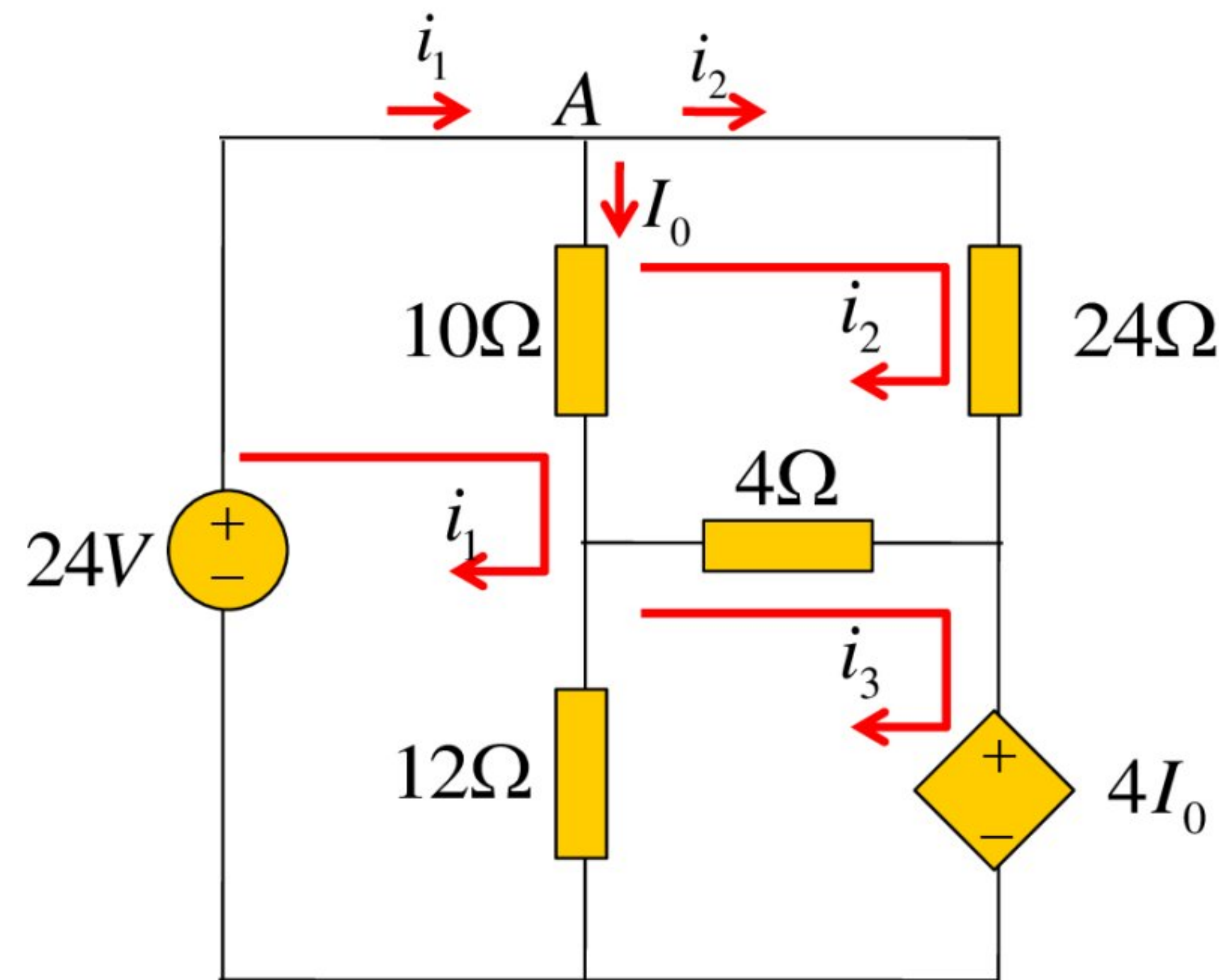
• تحليل الحلقات مع منابع التيار

Mesh Analysis with Current Sources

تحليل الحلقات

Example 3.6

باستخدام تحليل الحلقات اوجد التيار I_0 في الدارة التالية:



Solution - For mesh 1,

$$-24 + 10(i_1 - i_2) + 12(i_1 - i_3) = 0 \quad \text{or} \quad 11i_1 - 5i_2 - 6i_3 = 12 \quad (1)$$

تحليل الحلقات

- For mesh 2,

$$24i_2 + 4(i_2 - i_3) + 10(i_2 - i_1) = 0 \quad \text{or} \quad -5i_1 + 19i_2 - 2i_3 = 0 \quad (2)$$

- For mesh 3,

$$4I_0 + 12(i_3 - i_1) + 4(i_3 - i_2) = 0$$

But at node A, $I_0 = i_1 - i_2$, so that

$$4(i_1 - i_2) + 12(i_3 - i_1) + 4(i_3 - i_2) = 0 \quad \text{or} \quad -i_1 - i_2 + 2i_3 = 0 \quad (3)$$

In matrix form, equations 1, 2, and 3 become

$$\begin{bmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

تحليل الحلقات

We obtain the determinants as

$$\Delta = \begin{vmatrix} 11 & -5 & -6 \\ -5 & 19 & -2 \\ -1 & -1 & 2 \end{vmatrix} = 192, \quad \Delta_1 = \begin{vmatrix} 12 & -5 & -6 \\ 0 & 19 & -2 \\ 0 & -1 & 2 \end{vmatrix} = 432$$

$$\Delta_2 = \begin{vmatrix} 11 & 12 & -6 \\ -5 & 0 & -2 \\ -1 & 0 & 2 \end{vmatrix} = 144, \quad \Delta_3 = \begin{vmatrix} 11 & -5 & 12 \\ -5 & 19 & 0 \\ -1 & -1 & 0 \end{vmatrix} = 288.$$

We calculate the mesh currents using Cramer's rule as

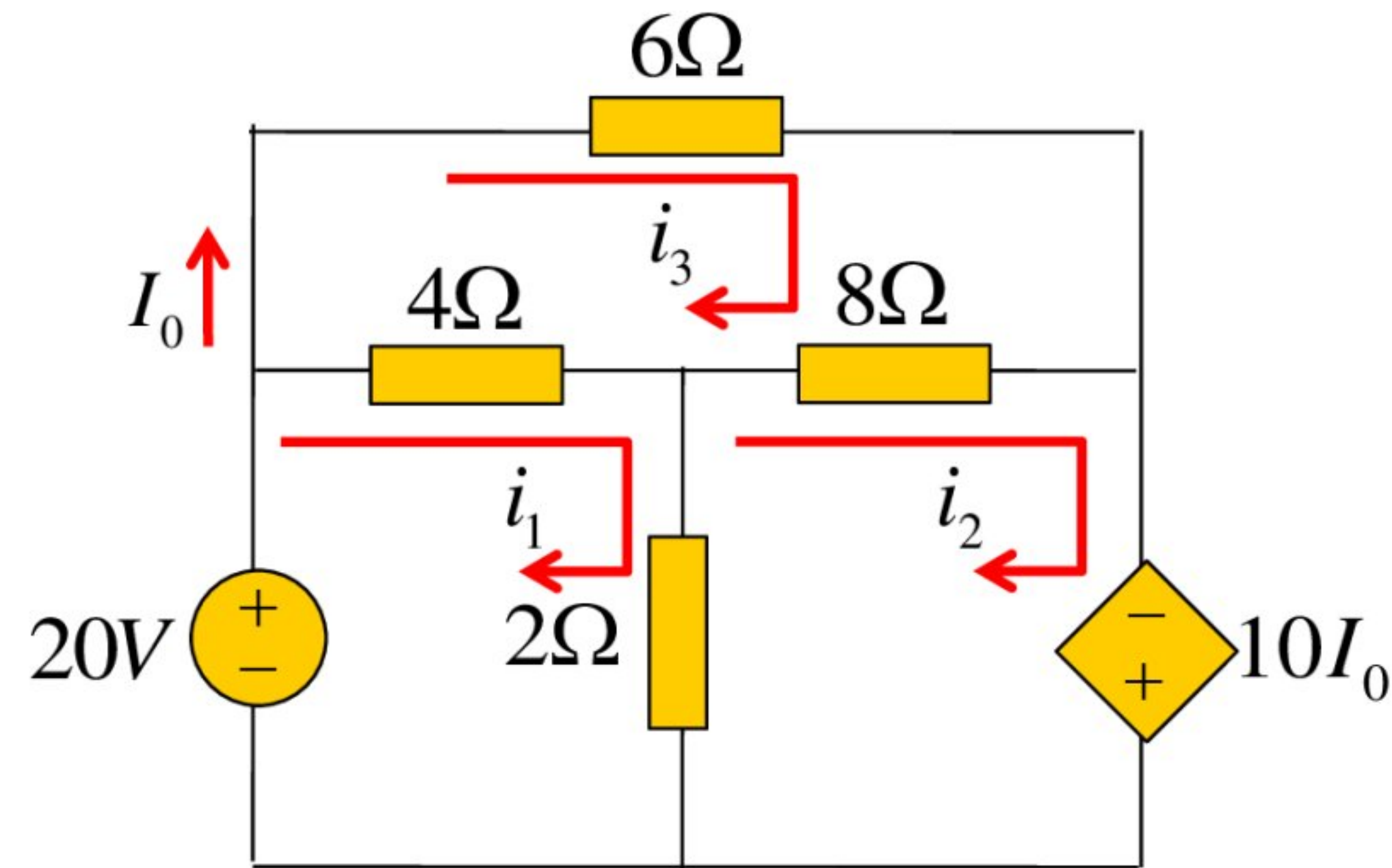
$$i_1 = \frac{\Delta_1}{\Delta} = 2.25 \text{ A}, \quad i_2 = \frac{\Delta_2}{\Delta} = 0.75 \text{ A}, \quad i_3 = \frac{\Delta_3}{\Delta} = 1.5 \text{ A}$$

Thus, $I_0 = i_1 - i_2 = 1.5 \text{ A}.$

تحليل الحلقات

Practice Problem 3.6

باستخدام تحليل الحلقات اوجد التيار I_0 في الدارة التالية:



Answer

$$I_0 = -5A.$$

Quiz

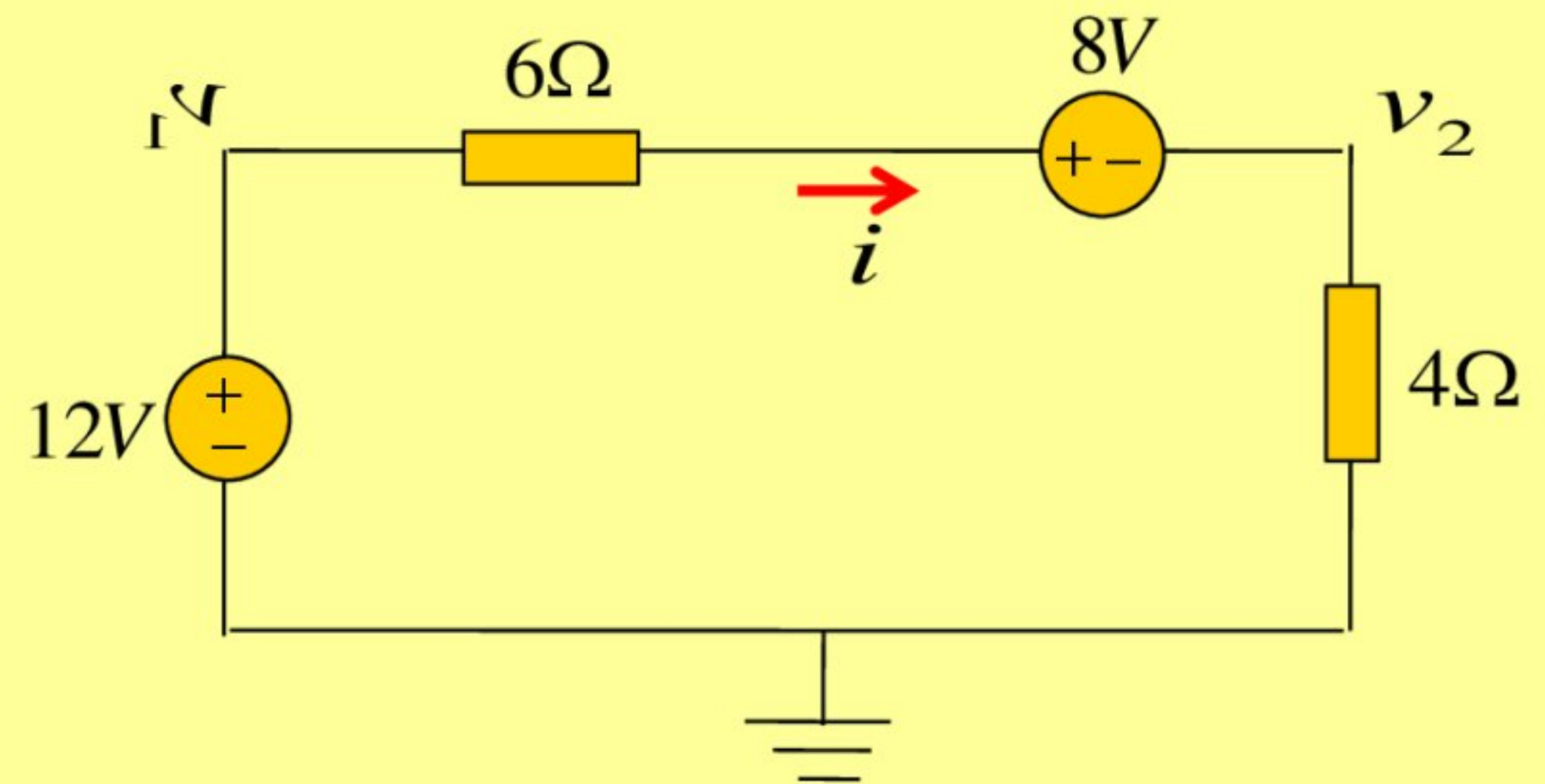
1. For the following circuit, v_1 and v_2 are related as:

a. $v_1 = 6i + 8 + v_2$

b. $v_1 = 6i - 8 + v_2$

c. $v_1 = -6i + 8 + v_2$

d. $v_1 = -6i - 8 + v_2$



2. In the above circuit, the voltage v_2 is:

a. -8 V

b. -1.6 V

c. 1.6 V

d. 8 V

Quiz

3. The current i in the following circuit is:

a. -2.667 A

b. -0.667 A

c. 0.667 A

d. 2.667 A

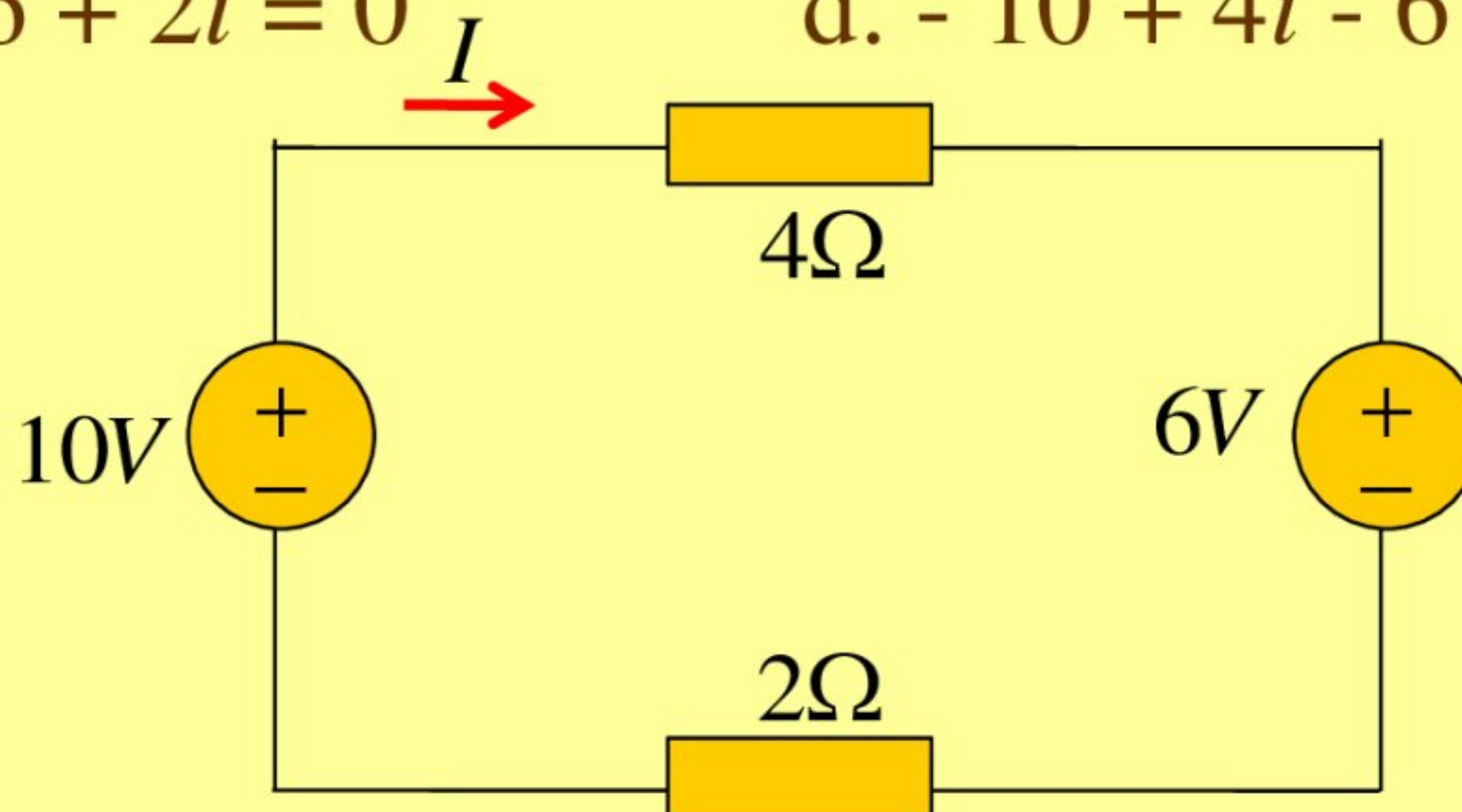
4. The loop equation for the following circuit is:

a. $-10 + 4i + 6 + 2i = 0$

b. $10 + 4i + 6 + 2i = 0$

c. $10 + 4i - 6 + 2i = 0$

d. $-10 + 4i - 6 + 2i = 0$



Quiz

5- In the following circuit, current i_1 is:

a. 4 A.

b. 3 A.

c. 2 A.

d. 1 A.

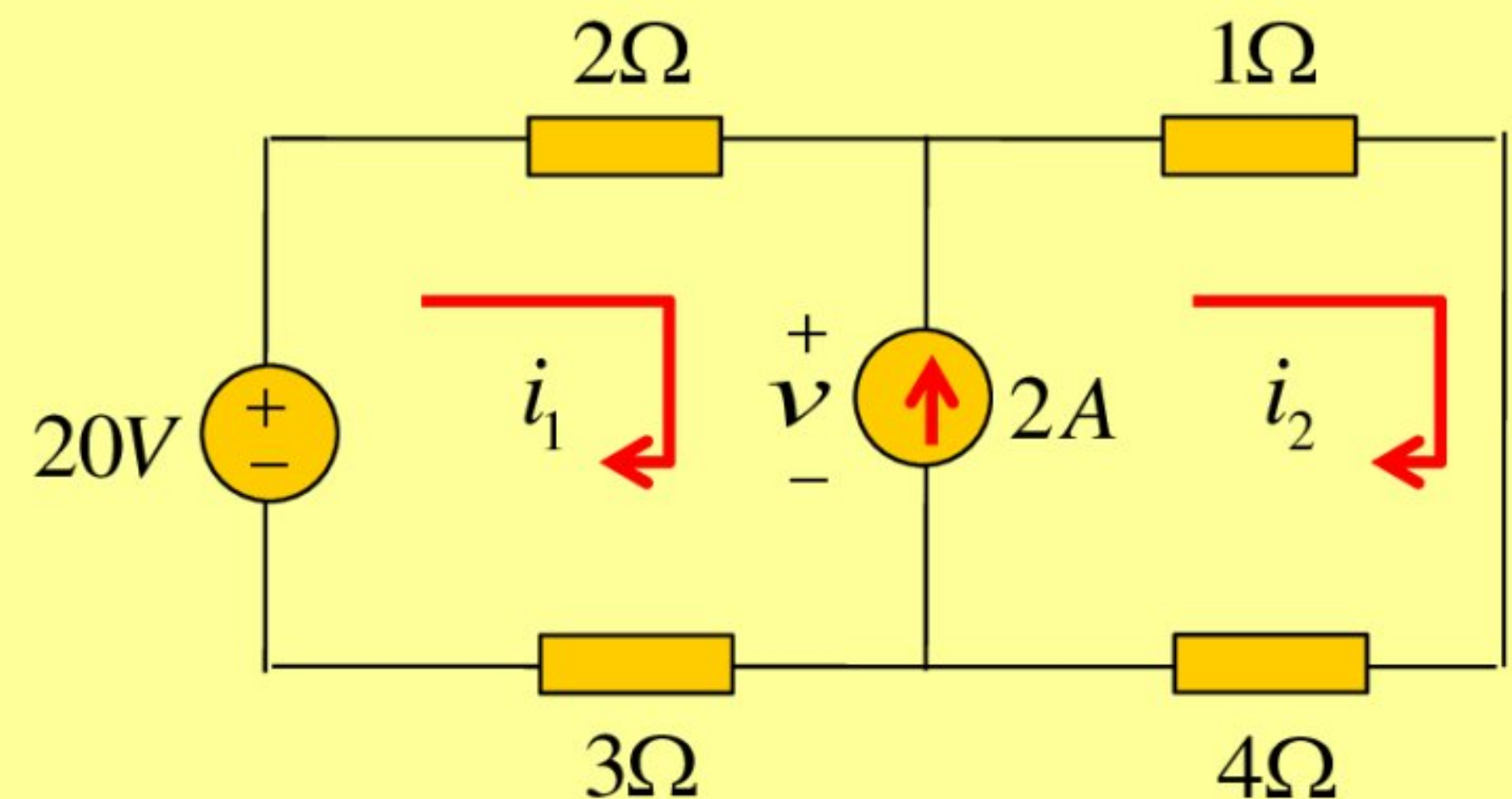
6- The voltage v across the current source in the following circuit is:

a. 20 V

b. 15 V

c. 10 V

d. 5 V



Quiz

Answers:

- | | |
|------|------|
| 1. a | 4. a |
| 2. c | 5. d |
| 3. c | 6. b |